



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY GUWAHATI

**Course Structure and Syllabus
(From Academic Session 2018-19 onwards)**

**B.TECH
ELECTRICAL AND ELECTRONICS ENGINEERING**

8th SEMESTER



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
Guwahati
Course Structure

(From Academic Session 2018-19 onwards)
B.Tech 8th Semester: Electrical and Electronics Engineering

Semester VIII/ B.TECH/EEE

Sl. No.	Sub-Code	Subject	Hours per Week			Credit	Marks	
			L	T	P	C	CE	ESE
Theory								
1	EEE1818PE5*	Program Elective-5	3	0	0	3	30	70
2	EEE1818PE6*	Program Elective-6	3	0	0	3	30	70
3	EEE1818PE7*	Program Elective-7	3	0	0	3	30	70
4	EEE1818OE4*	Open Elective-4	3	0	0	3	30	70
Practical								
1	EEE181821	Project -2	0	0	12	6	100	50
TOTAL			12	0	12	18	220	330
Total Contact Hours per week: 24								
Total Credits: 18								

Program Elective-5 Subjects		
Sl. No.	Subject Code	Subject
1	EEE1818PE51	Advanced Electric Drives
2	EE1818PE52	Smart Grid Technology
3	EEE1818PE5*	Any other subject offered from time to time with the approval of the University

Program Elective-6 Subjects		
Sl. No.	Subject Code	Subject
1	EE1818PE64	Industrial Electrical Systems
2	EEE1818PE61	Power System Protection
3	EEE1818PE6*	Any other subject offered from time to time with the approval of the University

Program Elective-7 Subjects		
Sl. No.	Subject Code	Subject
1	EEE1818PE71	High Voltage Engineering
2	EE1818PE73	Power System Instrumentation
3	EEE1818PE7*	Any other subject offered from time to time with the approval of the University

Open Elective-4 Subjects		
Sl. No.	Subject Code	Subject
1	EEE1818OE41	Hybrid Energy Systems
2	EEE1818OE42	FACTS
3	EEE1818OE4*	Any other subject offered from time to time with the approval of the University

Detailed Syllabus:

Course Code	Course Title	Hours per week L-T-P	Credit C
EEE1818PE51	Advanced Electric Drives	3-0-0	3

COURSE OBJECTIVES:

The main objective of this course is to make the students aware with the advanced application of power electronics in synchronous with the drives

COURSE OUTCOMES (CO): After completion of the course students will be able to:

CO1: Understand the operation of power electronic converters and their control strategies.

CO2: Understand the vector control strategies for ac motor drives

CO3: Understand the implementation of the control strategies using digital signal processors

MODULE 1: Power Converters for AC Drives

PWM control of inverter, harmonic elimination, space vector modulation, current control of VSI, three level inverter, Diode rectifier with boost chopper, PWM converter as line side rectifier, current fed inverters with self-commutated devices. Control of CSI, H bridge as a 4-Q drive

MODULE 2: Induction Motor Drives

Different transformations and reference frame theory, modeling of induction machines, voltage Fed inverter control-v/f control, vector control, direct torque and flux control (DTC)

MODULE 3: Synchronous Motor Drives

Modeling of synchronous machines, open loop v/f control, vector control, direct torque control, CSI fed synchronous motor drives.

MODULE 4: Permanent Magnet Motor Drives

Introduction to various PM motors, BLDC and PMSM drive configuration, comparison, block diagrams, Speed and torque control in BLDC and PMSM.

MODULE 5: Switched Reluctance Motor Drives

Evolution of switched reluctance motors, various topologies for SRM drives, comparison, Closed loop speed and torque control of SRM.

REFERENCES:

1. B. K. Bose, "Modern Power Electronics and AC Drives", Pearson Education, Asia, 2003.
2. P. C. Krause, O. Wasynczuk and S. D. Sudhoff, "Analysis of Electric Machinery and Drive Systems", John Wiley & Sons, 2013.
3. H. A. Taliyat and S. G. Campbell, "DSP based Electromechanical Motion Control", CRC press, 2003.
4. R. Krishnan, "Permanent Magnet Synchronous and Brushless DC motor Drives", CRC Press, 2009

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818PE52	Smart Grid Technology	3-0-0	3

COURSE OBJECTIVES:

To study distributed generation, smart transmission, smart distribution, smart consumption technology of electrical power, regulations and market models for smart grid

COURSE OUTCOMES (COs): After completion of the course students will be able to

CO1: describe structure, components and mathematical formulations for distributed generation, smart transmission, smart distribution, smart consumption technology of electrical power, regulations and market models for smart grid

CO2: solve problems of distributed generation, smart transmission, smart distribution, smart consumption technology of electrical power, regulations and market models for smart grid

CO3: analyze performance and characteristics of distributed generation, smart transmission, smart distribution, smart consumption technology of electrical power, regulations and market models for smart grid

MODULE 1: Introduction to Smart Grids

Definition, justification for smart grids, smart grid conceptual model, smart grid architectures, Interoperability, communication technologies, role of smart grids standards, intelligrid initiative, national smart grid mission (NSGM) by Govt. of India

MODULE 2: Smart Transmission Technologies

Substation automation, Supervisory control and data acquisition (SCADA), energy management system (EMS), phasor measurement units (PMU), Wide area measurement systems (WAMS)

MODULE 3: Smart Distribution Technologies

Distribution automation, outage management systems, automated meter reading (AMR), automated metering infrastructure (AMI), fault location isolation and service restoration (FLISR), Outage Management Systems (OMS), Energy Storage, Renewable Integration

MODULE 4: Distributed Generation and Smart Consumption

Distributed energy resources (DERs), smart appliances, low voltage DC (LVDC) distribution in homes or commercial buildings, home energy management system (HEMS), Net Metering, Building to Grid B2G, Vehicle to Grid V2G, Solar to Grid, Microgrid

MODULE 5: Regulations and Market Models for Smart Grid

Demand Response, Tariff Design, Time of the day pricing (TOD), Time of use pricing (TOU), Consumer privacy and data protection, consumer engagement; Cost benefit analysis of smart grid projects

TEXT BOOKS:

1. Clark W. Gellings: The Smart Grid, Enabling Energy Efficiency and Demand Side Response, CRC Press, 2009.
2. Jean Claude Sabonnadière, Nouredine Hadjsaïd: Smart Grids, Wiley-ISTE, IEEE Press, May 2012

REFERENCES:

1. Janaka Ekanayake, Kithsiri Liyanage, Jianzhong. Wu, Akihiko Yokoyama, Nick Jenkins: Smart Grid: Technology and Applications, Wiley, 2012.
2. James Momoh: Smart Grid: Fundamentals of Design and Analysis, Wiley, IEEE Press, 2012.
3. India Smart Grid Knowledge Portal

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818PE64	Industrial Electrical Systems	3-0-0	3

Course Outcomes: At the end of this course the student will be able to:

CO1: Explain the electrical wiring systems for residential, commercial and industrial consumers, representing the systems with standard symbols and drawings

CO2: Explain the construction and working components of industrial electrical systems.

CO3: Design the proper size of electrical system components.

MODULE 1: Electrical System Components

LT system wiring components, selection of cables, wires, switches, distribution box, metering system, Tariff structure, protection components- Fuse, MCB, MCCB, ELCB, inverse current characteristics, symbols, single line diagram (SLD) of a wiring system, Contactor, Isolator, Relays, MPCB, Electric shock and Electrical safety practices

MODULE 2: Residential and Commercial Electrical Systems

Types of residential and commercial wiring systems, general rules and guidelines for installation, load calculation and sizing of wire, rating of main switch, distribution board and protection devices, earthing system calculations, requirements of commercial installation, deciding lighting scheme and number of lamps, earthing of commercial installation, selection and sizing of components.

MODULE 3: Illumination Systems

Understanding various terms regarding light, lumen, intensity, candle power, lamp efficiency, specific consumption, glare, space to height ratio, waste light factor, depreciation factor, various illumination schemes, Incandescent lamps and modern luminaries like CFL, LED and their operation, energy saving in illumination systems, design of a lighting scheme for a residential and commercial premises, flood lighting.

MODULE 4: Industrial Electrical Systems-I

High Tension (HT) connection, industrial substation, Transformer selection, Industrial loads, motors, starting of motors, Single Line Diagram (SLD), Cable and Switchgear selection, Lightning Protection, Earthing design, Power factor correction – kVAR calculations, type of compensation, Introduction to PCC, MCC panels; Specifications of Low Tension (LT) Breakers, Miniature Circuit Breaker (MCB) and other LT panel components.

MODULE 5: Industrial Electrical Systems-II

Diesel Generator (DG) Systems, UPS System, Electrical Systems for the elevators, Battery banks, Sizing the DG, Uninterrupted Power Supply (UPS) and Battery Banks, Selection of UPS and Battery Banks

MODULE 6: Industrial Electrical System Automation

Study of basic Programmable Logic Controller (PLC), Role of in automation, advantages of process automation, PLC based control system design, Panel Metering and Introduction to SCADA system for distribution automation.

Text Books:

1. S. L. Uppal and G. C. Garg, -Electrical Wiring, Estimating & Costing, Khanna publishers, 2008.
2. H. Joshi, -Residential, Commercial and Industrial Systems, McGraw-Hill, 2008

References:

1. K. B. Raina, -Electrical Design, Estimating & Costing, New age International, 2007
2. S. Singh and R. D. Singh, -Electrical estimating and costing, Dhanpat Rai and Co., 1997
3. Web site for IS Standards.

Course Code	Course Title	Hours per week L-T-P	Credit C
EEE1818PE61	Power System Protection	3-0-0	3

Course Outcomes (CO): At the end of this course, students will demonstrate the ability to

CO1: Understand the different components of a protection system.

CO2: Evaluate fault current due to different types of faults in a network.

CO3: Understand the protection schemes for different power system components.

CO4: Understand the basic principles of digital protection.

CO5: Understand system protection schemes, and the use of wide-area measurements.

MODULE 1: Introduction and Components of a Protection System

Principles of Power System Protection, Relays, Instrument transformers, Circuit Breakers

MODULE 2: Faults and Over-Current Protection

Review of Fault Analysis, Sequence Networks. Introduction to Overcurrent Protection and overcurrent relay co-ordination.

MODULE 3: Equipment Protection Schemes

Directional, Distance, Differential protection. Transformer and Generator protection. Bus bar Protection, Bus Bar arrangement schemes.

MODULE 4: Modeling and Simulation of Protection Schemes

CT/PT modeling and standards, Simulation of transients using Electro-Magnetic Transients (EMT) programs. Relay Testing.

MODULE 5: System Protection

Effect of Power Swings on Distance Relaying. System Protection Schemes. Under-frequency, under-voltage and df/dt relays, Out-of-step protection, Synchro-phasors, Phasor Measurement Units and Wide-Area Measurement Systems (WAMS). Application of WAMS for improving protection systems.

REFERENCES:

1. J. L. Blackburn, "Protective Relaying: Principles and Applications", Marcel Dekker, New York, 1987.
2. Y. G. Paithankar and S. R. Bhide, "Fundamentals of power system protection", Prentice Hall, India, 2010.
3. A. G. Phadke and J. S. Thorp, "Computer Relaying for Power Systems", John Wiley & Sons, 1988.
4. A. G. Phadke and J. S. Thorp, "Synchronized Phasor Measurements and their Applications", Springer, 2008.
5. D. Reimert, "Protective Relaying for Power Generation Systems", Taylor and Francis

Course Code	Course Title	Hours per week L-T-P	Credit C
EEE1818PE71	High Voltage Engineering	3-0-0	3

Course Outcomes (CO): At the end of the course, the students will be able to

CO1: Understand the basic physics related to various breakdown processes in solid, liquid and gaseous insulating materials.

CO2: Knowledge of generation and measurement of D. C., A.C., & Impulse voltages.

CO3: Knowledge of tests on H. V. equipment and on insulating materials, as per the standards.

CO4: Knowledge of how over-voltages arise in a power system, and protection against these over-voltages.

MODULE 1: Breakdown in Gases

Ionization processes and de-ionization processes, Types of Discharge, Gases as insulating materials, Breakdown in Uniform gap, non-uniform gaps, Townsend's theory, Streamer mechanism, Corona discharge

MODULE 2: Breakdown in Liquid and Solid Insulating Materials

Breakdown in pure and commercial liquids, Solid dielectrics and composite dielectrics, intrinsic breakdown, electromechanical breakdown and thermal breakdown, Partial discharge, applications of insulating materials.

MODULE 3: Generation of High Voltages

Generation of high voltages, generation of high D. C. and A.C. voltages, generation of impulse voltages, generation of impulse currents, tripping and control of impulse generators.

MODULE 4: Measurements of High Voltages and Currents

Peak voltage, impulse voltage and high direct current measurement method, cathode ray oscillographs for impulse voltage and current measurement, measurement of dielectric constant and loss factor, partial discharge measurements.

MODULE 5: Lightning and Switching Over-voltages

Charge formation in clouds, Stepped leader, Dart leader, Lightning Surges. Switching over- voltages, Protection against over-voltages, Surge diverters, Surge modifiers.

REFERENCES:

1. M. S. Naidu and V. Kamaraju, "High Voltage Engineering", McGraw Hill Education, 2013.
2. C. L. Wadhwa, "High Voltage Engineering", New Age International Publishers, 2007.

Course Code	Course Title	Hours per week L-T-P	Credit C
EE1818PE73	Power System Instrumentation	3-0-0	3

MODULE 1: Different Types of Instruments

Voltmeter–analog /digital, ammeter-analog/digital, frequency meter, induction type energy meter, 3 phase energy meter, Vector meter, tri vector meter, analog VAR meter.

MODULE 2: Measurement of Large Currents and Voltages

introduction, high voltage measurement technique for DC/AC voltage measurement, Buffer system: ratio and phase angle error, Pulse trained energy meter

MODULE 3: Measurements in Power Plants

Electrical measurements – Current, voltage, power, frequency, power factor etc. – Nonelectrical parameters – Flow of feed water, fuel, air and steam with correction factor for temperature – Steam pressure and steam temperature – Drum level measurement – Radiation detector – Smoke density measurement – Dust monitor

MODULE 4: Data Loggers and Data Display System

MODULE 5: Telemetry

Introduction, component, short term telemetering, Data transmission channels-PLCC, microwave links

MODULE 6: SCADA

MODULE 7: Introduction to PMUs and their Placement

REFERENCES:

1. Cooper Helfrick, “Electrical Instrumentation and Measuring Techniques”, Prentice Hall India, 1986
2. D. C. Nakra and K. K. Chowdhry, “Instrumentation, Measurement, and Analysis”, Tata McGraw Hill Publishing Co., 1984.
3. Selected topics from IEEE, AIEE and CIGRE Journals

Course Code	Course Title	Hours per week L-T-P	Credit C
EEE1818OE41	Hybrid Energy Systems	3-0-0	3

Course Outcomes (CO): At the end of this course, students will demonstrate the ability to

CO1: Understand the energy scenario and the consequent growth of the power generation from renewable energy sources.

CO2: Understand the basic physics of wind and solar power generation.

CO3: Understand the power electronic interfaces for wind and solar generation.

CO4: Understand the issues related to the grid-integration of solar and wind energy systems.

MODULE 1: Physics of Wind Power

History of wind power, Indian and Global statistics, Wind physics, Betz limit, Tip speed ratio, stall and pitch control.

MODULE 2: Wind Generator Topologies

Review of modern wind turbine technologies, Fixed and Variable speed wind turbines, Induction Generators, Doubly-Fed Induction Generators and their characteristics, Permanent- Magnet Synchronous Generators, Power electronics converters.

MODULE 3: The Solar Resource

Introduction, solar radiation spectra, solar geometry, Earth Sun angles, observer Sun angles, solar day length, Estimation of solar energy availability.

MODULE 4: Solar Photovoltaic

Technologies-Amorphous, monocrystalline, polycrystalline; V-I characteristics of a PV cell, PV module, array, Power Electronic Converters for Solar Systems, Maximum Power Point Tracking (MPPT) algorithms.

MODULE 5: Network Integration Issues

Overview of grid code technical requirements. Fault ride-through for wind farms - real and reactive power regulation, voltage and frequency operating limits, solar PV and wind farm behavior during grid disturbances. Power quality issues. Power system interconnection experiences in the world. Hybrid and isolated operations of solar PV and wind systems.

REFERENCES:

1. T. Ackermann, "Wind Power in Power Systems", John Wiley and Sons Ltd., 2005.
2. G. M. Masters, "Renewable and Efficient Electric Power Systems", John Wiley and Sons, 2004.
3. S. P. Sukhatme, "Solar Energy: Principles of Thermal Collection and Storage", McGraw Hill, 1984.
4. H. Siegfried and R. Waddington, "Grid integration of wind energy conversion systems" John Wiley and Sons Ltd., 2006.
5. G. N. Tiwari and M. K. Ghosal, "Renewable Energy Applications", Narosa Publications, 2004.
6. J. A. Duffie and W. A. Beckman, "Solar Engineering of Thermal Processes", John Wiley & Sons, 1991.

Course Code	Course Title	Hours per week L-T-P	Credit C
EEE1818OE42	FACTS	3-0-0	3

Course Outcomes (CO): At the end of this course, students will demonstrate the ability to

CO1: Understand the characteristics of ac transmission and the effect of shunt and series reactive compensation.

CO2: Understand the working principles of FACTS devices and their operating characteristics.

CO3: Understand the basic concepts of power quality.

CO4: Understand the working principles of devices to improve power quality.

MODULE 1: Transmission Lines and Series/Shunt Reactive Power Compensation

Basics of AC Transmission. Analysis of uncompensated AC transmission lines. Passive Reactive Power Compensation. Shunt and series compensation at the mid-point of an AC Line. Comparison of Series and Shunt Compensation.

MODULE 2: Thyristor-Based Flexible AC Transmission Controllers (FACTS)

Description and Characteristics of Thyristor-based FACTS devices: Static VAR Compensator (SVC), Thyristor Controlled Series Capacitor (TCSC), Configurations/Modes of Operation, Harmonics and control of SVC and TCSC. Fault Current Limiter.

MODULE 3: Voltage Source Converter Based (FACTS) Controllers

Voltage Source Converters (VSC): Six Pulse VSC, Multi-pulse, STATCOM: Principle of Operation, Static Synchronous Series Compensator (SSSC) and Unified Power Flow Controller (UPFC): Principle of Operation and Control. GTO Controlled Series Compensator. Fault Current Limiter.

MODULE 4: Application of FACTS

Application of FACTS devices for power-flow control and stability improvement. Simulation of power swing damping in a single-machine infinite bus system using a TCSC. Simulation example of voltage regulation of transmission mid-point voltage using a STATCOM.

MODULE 5: Power Quality Problems in Distribution Systems

Power Quality problems in distribution systems: Transient and Steady state variations in voltage and frequency. Unbalance, Sags, Swells, Wave-form Distortions: harmonics, noise, notching, dc-offsets, fluctuations. Flicker and its measurement. Tolerance of Equipment

REFERENCES:

1. N. G. Hingorani and L. Gyugyi, "Understanding FACTS: Concepts and Technology of FACTS Systems", Wiley-IEEE Press, 1999.
2. K. R. Padiyar, "FACTS Controllers in Power Transmission and Distribution", New Age International (P) Ltd. 2007.
3. T. J. E. Miller, "Reactive Power Control in Electric Systems", John Wiley and Sons, New York, 1983.
4. R. C. Dugan, "Electrical Power Systems Quality", McGraw Hill Education, 2012.
